



Mohammed HAOUACH

AI/DATA ENGINEER

Skills

DATA

- **Conceptual management of data pipelines (ETL/ELT)** data (batch and real-time)
- **Integration** (batch and real-time)
- **Database performance optimization**
- **Data modeling (OLAP, OLTP, Data Lake, Data Warehouse)**
- **Big Data, Graph database, Vector database**

AI/ML

- **Prétraitement et transformation de données pour le ML**
- **Model deployment (MLOps)**
- **Agentic AI, MCP, RAG, Langchain, N8N automations**
- **Forecasting and anomaly detection of time series**
- **Evaluation and monitoring of models in production**

TECHNOS

- **Python: Pandas, NumPy, PyTorch, TensorFlow, Scikit-learn**
- **Big Data: Apache Spark, Hadoop, Kafka, Apache Airflow, Hadoop, Kafka, Apache Airflow**
- **Databases: PostgreSQL, MongoDB, Teradata, BigQuery, Neo4j Databases: PostgreSQL, MongoDB, Teradata, BigQuery, Neo4j**
- **AI NLP: RAG, Fine-tuning, Langchain, Langfuse, Pandas AI, MCP**
- **MLOps/Services: MLflow, DVC, Docker, FastAPI, Flask, Django**
- **Cloud: Azure, AWS (S3, Redshift, EMR), GCP (Vertex AI, Redshift, EMR), GCP (Vertex AI)**
- **CI/CD: Git, GitLab CI/CD, Jenkins, Harbor, Docker, Kubernetes**

API MONITORING MONITORING

- **Django, FastAPI, Flask, Dash/Plotly, PowerBI and Tableau**
- **Tracking model performance with Prometheus/Grafana**

Professional experience



DATA SCIENTIST CNAV Since Sept 2024

- **Optimization of the energy consumption of a data center** by detecting unused VMs.
- **Pipelines ETL pour agréger des données d'API internes.**
- **Design of AI models for time series embedding** with LSTM and Transformers
- **Clustering to detect similar behaviors** similar



IT PROJECT MANAGER SAMA CONCEPT April 2013 - Oct 2023

- **Management of a computer vision project applied to quality control on a production line.**
- **Mise en œuvre de pipelines de traitement d'image temps réel** for industrial inspection



RESEARCH ENGINEER CHANEL

Sept 2007 - Fév 2013
Feb 2013

- **Design and development of 3D reconstruction software** of the face from stereo images, segmentation of pathologies skin and calculation of wrinkle depth.
- **Development of an interactive cube visualization tool** 3D OLAP, with cluster extraction

Training

DOCTORANT MACHINE LEARNING 2007-2010

Université François Rabelais de Tours
University of Tours
Subject: Acquisition, annotation and interactive extraction of images stereoscopic in virtual reality

MASTER 2 MACHINE LEARNING 2005-2006

Polytech'Tours
Intelligent systems and applications

COMPUTER ENGINEERING 2000-2005

School of Higher Studies in Computer Science
• **Data Analysis - Decision Informatics**

Contact details

Address:

8 rue Pierre de Coubertin, 37300

Played Tours

Email:

haouach.mohammed@gmail.com

Tel.

06 09 15 39 30

Certifications

Developer AI:

Professional certification

RNCP 3927

Azure Cloud:

Microsoft Azure AI

Microsoft Azure Cloud

Microsoft MCP:

Windows Server 070-810-210

CISCO CCNA & CCNP:

Industrial 640-801801

Languages

French: Native language

50) English: Good level (B2)

5th) Arabic: Good level (B2)